Logistics Hub Performance Analysis

Operational Efficiency & Delivery Delay Drivers

Tools: Excel | SQL | Power BI

J.L. Coloma | Data Analyst

Analysis Objectives

01

Hub Operational Performance

- Monitor hub operations from arrival to dispatch
- Identify **bottlenecks** in yard waiting, dock operations and turnaround time

02

External Delay Factors

- Analyze the **impact of traffic, weather** and geographic area
- Measure their effect on delivery time and on-time rate
- Improve performance through data insights

Key Questions:

- Does peak demand reduce operational performance?
- Which external factors impact delivery time the most?
- Does a delayed truck arrival impact the final delivery time?

Dataset

Amazon Delivery Dataset (Kaggle) | 5,000 orders

Analysis Workflow

1. **Excel:** Dataset cleaning & feature engineering
2. **SQL:** Aggregations, CASE WHEN, KPI calculations
3. **Power BI:** Dashboard & KPI visualization

Core KPIs

- Avg Delivery Time
- On-Time Delivery Rate (%)
- Hub Turnaround Time
- Yard Waiting Time

Feature Engineering in Excel

- **Hour**

  Extracted from Order_Time

- **Shift**

  IF(Hour ≥ 22 OR Hour < 6 → Night, else Day)

- **Yard_Wait**

  Dock_Entry Time - Truck_Arrival Time

- **Traffic_Level**

  Numeric encoding of Traffic_Condition

Order_ID	Hour	Shift	Truck_Arrival	Dock_Entry	Hub_Exit	Delivery_Time	Status	**Yard_Wait**	**Turnaround**
ialx566343618	11	Day	12:23	12:39	13:13	120	On Time	16	50
akqg208421122	19	Day	20:11	20:33	20:50	165	Late	22	39
njpu434582536	8	Day	09:10	09:26	09:51	130	Late	16	41
rjto796129700	18	Day	18:43	18:54	19:37	105	On Time	11	54
zguw716275638	13	Day	13:56	14:19	14:53	150	Late	23	57

Data prepared and structured for operational KPI analysis

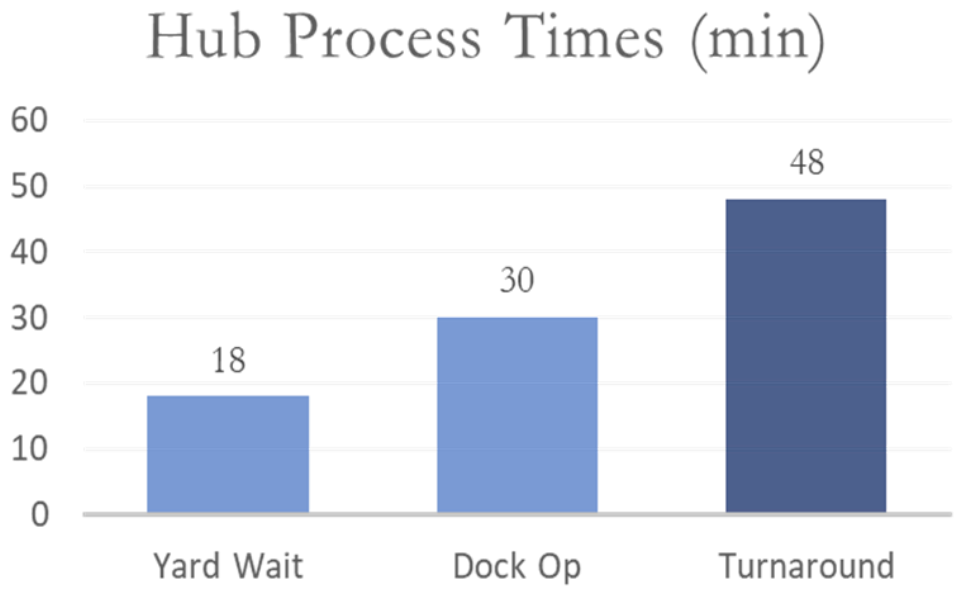
Hub Operations Are Not the Source of Delivery Delays

- Hub performance remains stable across all hours
- No internal bottlenecks detected (yard, dock, turnaround)

→ Delivery delays are driven by external factors

Order Volume by Hour

Hour	Orders
9 (Off-Peak)	222
14 (Mid)	83
19 (Peak)	519



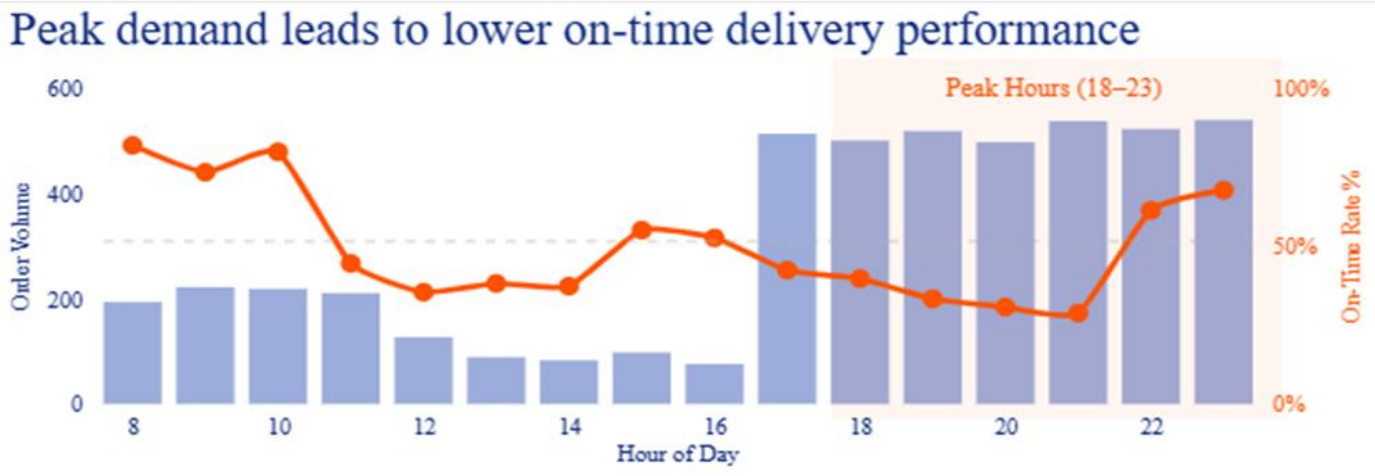
Turnaround time remains stable (~48 min) across all hours

SQL Validation

```
SELECT
    hour,
    ROUND(AVG(yard_waiting_time_min),0) AS avg_yard_wait,
    ROUND(AVG(dock_operation_time_min),0) AS avg_dock_time,
    ROUND(AVG(hub_turnaround_time_min),0) AS avg_turnaround
FROM deliveries
GROUP BY hour
ORDER BY hour;
```

Demand Does Not Fully Explain Delivery Delays

Mid-day collapse (11:00–14:00): on-time rate drops from ~**80%** to ~**35%** despite low demand — confirming delays are driven by external factors, not hub operations



- Demand peaks between 18:00–23:00
- On-time performance decreases during peak hours

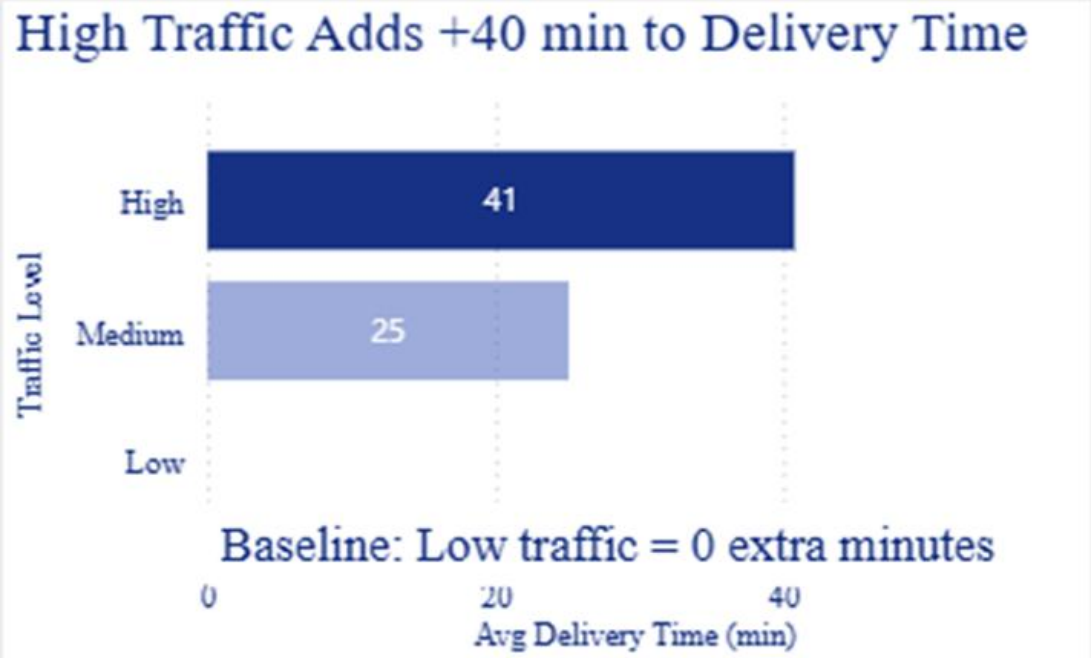
→ Mid-day drop confirms delays are not driven by demand alone

Traffic Is the Primary Driver of Delivery Delays

High traffic: ~**30%** on-time | **+41 min** vs low traffic

- High traffic adds +41 min to delivery time
- On-time rate drops from 71% → 32%
- Traffic explains performance drops even at low demand

→ Traffic is the main operational constraint impacting delivery performance

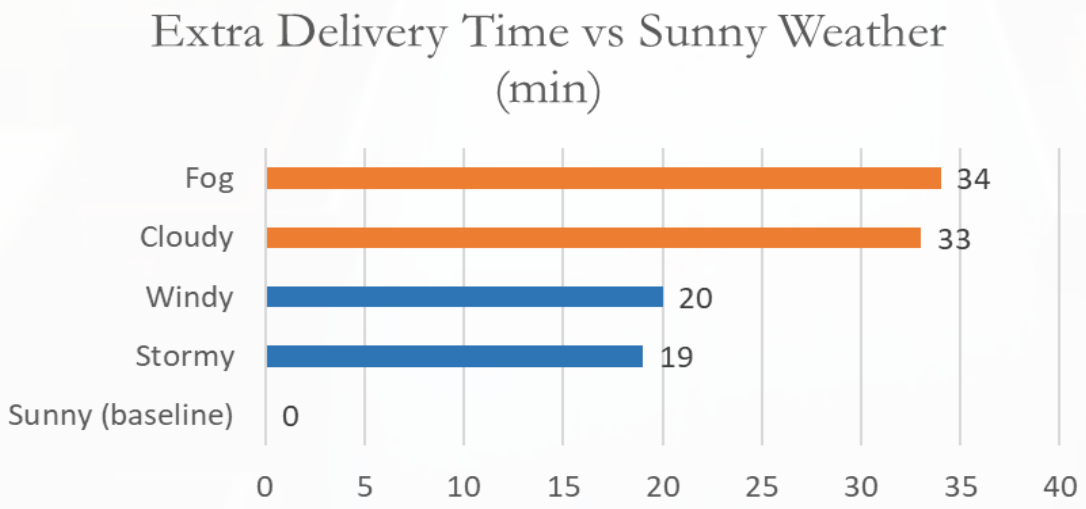


SQL Validation

```
SELECT traffic_condition,
  ROUND(AVG(delivery_time_min), 0) AS avg_delivery_time,
  ROUND(AVG(delivery_time_min)
          - MIN(AVG(delivery_time_min)) OVER (), 0)
  AS extra_time_vs_low,
  ROUND(
     SUM(CASE WHEN on_time_status = 'On Time’
     THEN 1 ELSE 0 END) * 100.0
     / COUNT(order_id), 0
     ) AS on_time_rate_pct
FROM deliveries
GROUP BY traffic_condition
ORDER BY avg_delivery_time DESC;
```

Fog Drives Higher Delays Than Stormy Weather

Fog & Cloudy: +33–34 min (highest impact)
Storm: +19 min, lower than expected



- Fog and cloudy conditions drive the highest delays
- Storm impact is lower than expected

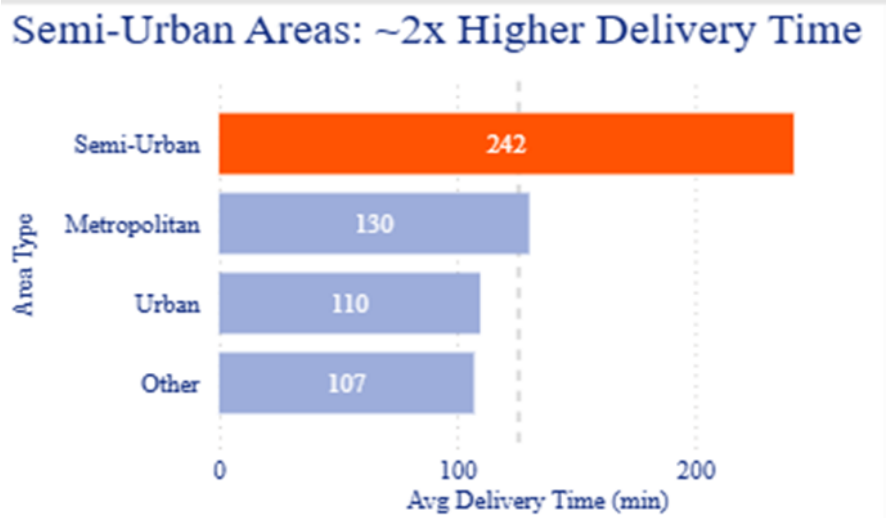
→ Secondary factor compared to traffic

→ Low visibility is the key weather-related constraint

Geographic Area Is a Key Driver of Delivery Performance Differences

- Semi-urban areas show the highest delivery times (~2x)
- Urban and metropolitan areas significantly outperform semi-urban areas

→ Geographic area drives performance differences



Vehicle Performance by Area

Derived using SQL aggregations

Area	Vehicle	Avg Time	On-Time %
Metropolitan	Motorcycle	136 min	40%
Metropolitan	Scooter	122 min	49%
Urban	Motorcycle	116 min	61%
Urban	Scooter	103 min	67%
Other	Scooter	101 min	71%

Vehicle type is not a primary driver

→ Performance is driven by geographic area

Combined External Factors Drive Extreme Delivery Delays

➢ Worst-case scenario:High traffic + fog
→ up to 170 min delivery time
→ Only ~10–12% on-time performance

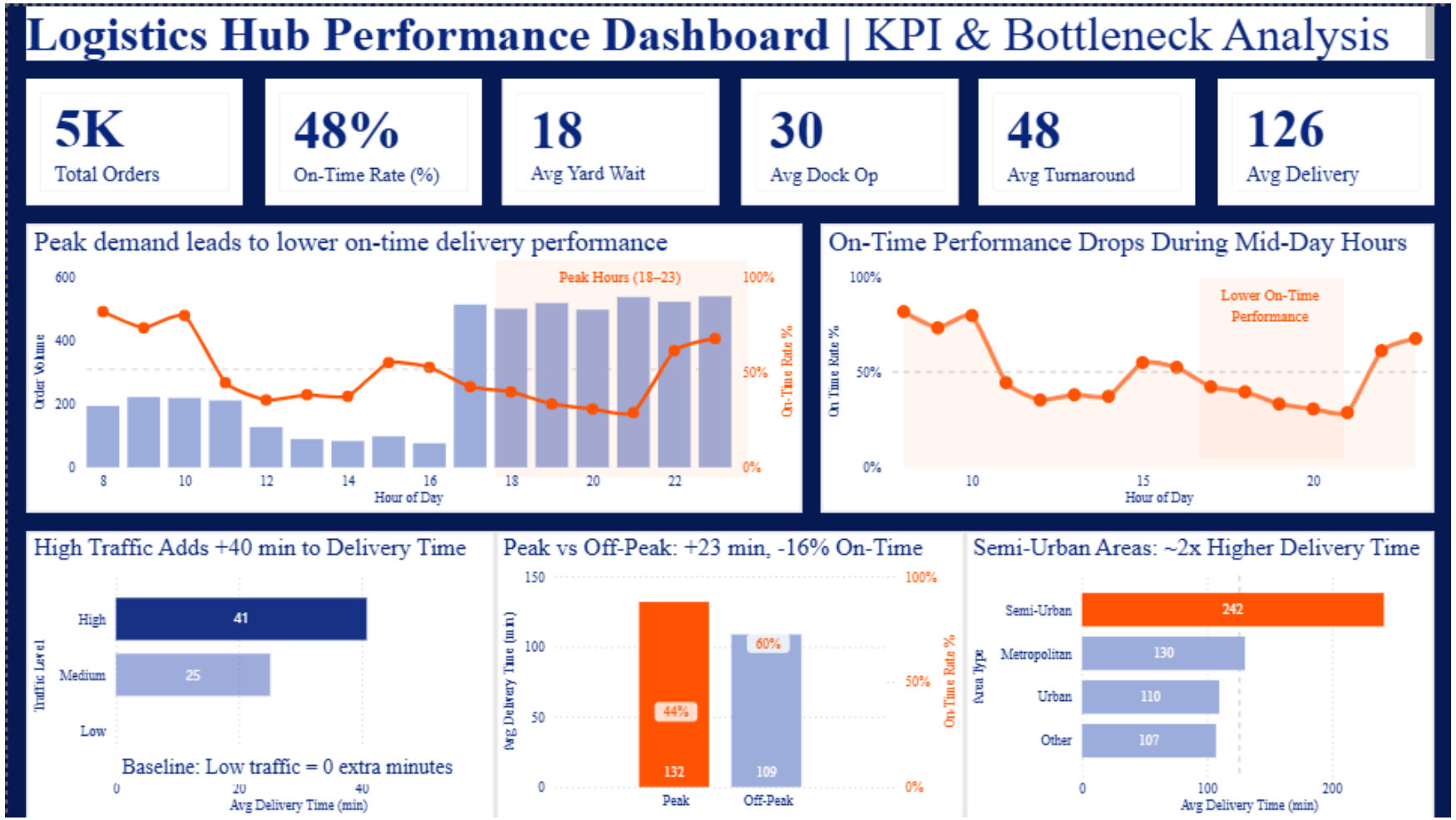
Area	Traffic	Weather	Avg Time	On-Time %	Risk
Metropolitan	High	Fog	170 min	12%	🔴 Critical
Urban	High	Cloudy	169 min	10%	🔴 Critical
Metropolitan	High	Cloudy	163 min	14%	🔴 Critical
Urban	High	Fog	160 min	21%	🔴 Critical
Metropolitan	Medium	Fog	146 min	28%	⚠️ Warning

→ Delivery performance collapses when external risk factors overlap

SQL Validation

```
SELECT
    area_type,
    traffic_condition,
    weather_condition,
    ROUND(AVG(delivery_time_min), 0) AS avg_time,
    ROUND(
        SUM(CASE WHEN on_time_status = 'On Time' THEN 1 ELSE 0 END) * 100.0
        / COUNT(order_id), 0
    ) AS on_time_rate
FROM deliveries
GROUP BY area_type, traffic_condition, weather_condition
HAVING AVG(delivery_time_min) > 130
ORDER BY avg_time DESC;
```

Power BI Dashboard: From Raw Data to Actionable Insights



- Tracks delivery performance across time, traffic and geographic areas
- Compares peak vs off-peak conditions to identify performance drops
- Monitors hub operations (yard, dock, turnaround) to confirm internal efficiency

Built in Power BI | Data source: PostgreSQL via SQL queries

Conclusions & Recommendations

KEY FINDINGS

Hub operations are not a bottleneck
Turnaround remains stable (~47–48 min) across all conditions

Delivery performance is driven by external factors
Traffic is the main driver (+40%), followed by weather (+34 min) and geography

Semi-urban areas represent the primary geographic bottleneck (~2x delivery time)
Significant performance gap vs urban and metropolitan areas

RECOMMENDATIONS

Optimize routing based on traffic conditions
Prioritize low-traffic routes and avoid peak congestion

Implement weather-aware scheduling
Limit time-sensitive deliveries during low visibility conditions

Rebalance operations in semi-urban areas
Optimize routing and adjust resource allocation to reduce inefficiencies

→ Improving delivery performance requires addressing external factors